Activity 4— Campus Event Financial Planning

1. Problem Statement

Develop a Python program to calculate the total expenses and remaining budget for a college event. The program takes the event budget and different expenses as input and displays the total expenses and remaining amount.

2. Objectives

- To calculate the total expenses of a college event.
- To compare expenses with the available budget.
- To calculate the remaining budget.
- To understand Python variables, arithmetic operators and conditional statements.

3. Assumptions

For this example, assume:

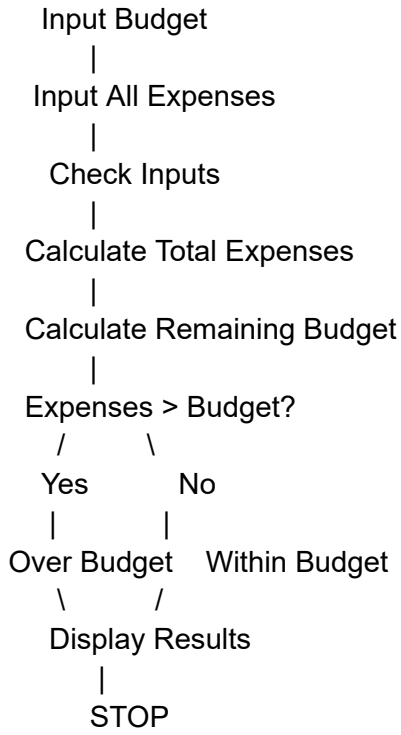
- Total event budget = ₹10,000
- Venue expenses, food expenses and decoration expenses are entered by the user.
- All expenses must be zero or positive.
- The program checks whether the event is within budget.

4. Algorithm

1. Start.
2. Enter the total event budget.
3. Enter venue, food and decoration expenses.
4. Check whether all inputs are valid.
5. Calculate total expenses.
6. Calculate the remaining budget.
7. Check whether expenses exceed the budget.
8. Display the total expenses and budget status.
9. Stop.

5. Flowchart

START
|



6. Pseudocode

```
START
INPUT budget
INPUT venue
INPUT food
INPUT decoration

IF any input is negative
   PRINT "Invalid input"
ELSE
   total = venue + food + decoration
   remaining = budget - total

   PRINT total
   PRINT remaining

   IF total > budget
      PRINT "Over Budget"
   ELSE
      PRINT "Within Budget"
   ENDIF
ENDIF

STOP
```

7. Python Source Code

File name: `activity4.py`

```
budget = float(input("Enter event budget: "))
venue = float(input("Enter venue expenses: "))
food = float(input("Enter food expenses: "))
decoration = float(input("Enter decoration expenses: "))

if budget < 0 or venue < 0 or food < 0 or decoration < 0:
    print("Invalid input")

else:
    total = venue + food + decoration
    remaining = budget - total

    print("Total Expenses: Rs.", total)
    print("Remaining Budget: Rs.", remaining)

    if total > budget:
        print("Status: Over Budget")
    else:
        print("Status: Within Budget")
```

8. Test Cases

Budget	Venue	Food	Decoration	Expected Result
₹10,000	₹2,000	₹3,000	₹1,000	₹4,000 remaining
₹10,000	₹4,000	₹4,000	₹3,000	₹1,000 over budget
₹5,000	₹1,000	₹1,000	₹1,000	₹2,000 remaining
-₹5,000	₹1,000	₹1,000	₹500	Invalid input

9. Output Screenshot and Conclusion

Sample Output:

```
Enter event budget: 10000
Enter venue expenses: 2000
Enter food expenses: 3000
Enter decoration expenses: 1000
Total Expenses: Rs. 6000.0
Remaining Budget: Rs. 4000.0
Status: Within Budget
```

Output Screenshot: Run `activity4.py` in Python IDLE or VS Code. Capture the inputs and output, and save the screenshot as `activity4_output.png`.

Conclusion: This program calculates college event expenses and checks whether they are within the available budget using Python arithmetic and conditional statements.

Input code:

```
budget = float(input("Enter event budget: "))
venue = float(input("Enter venue expenses: "))
food = float(input("Enter food expenses: "))
decoration = float(input("Enter decoration expenses: "))

if budget < 0 or venue < 0 or food < 0 or decoration < 0:
    print("Invalid input")

else:
    total = venue + food + decoration
    remaining = budget - total

    print("Total Expenses: Rs.", total)
    print("Remaining Budget: Rs.", remaining)

    if total > budget:
        print("Status: Over Budget")
    else:
        print("Status: Within Budget")
```

output:

```
Enter event budget: 10000
Enter venue expenses: 2000
Enter food expenses: 3000
Enter decoration expenses: 1000
Total Expenses: Rs. 6000.0
Remaining Budget: Rs. 4000.0
Status: Within Budget

==================================== REST
Enter event budget: 10000
Enter venue expenses: 4000
Enter food expenses: 4000
Enter decoration expenses: 3000
Total Expenses: Rs. 11000.0
Remaining Budget: Rs. -1000.0
Status: Over Budget

==================================== REST
Enter event budget: 5000
Enter venue expenses: 1000
Enter food expenses: 1000
Enter decoration expenses: 1000
Total Expenses: Rs. 3000.0
Remaining Budget: Rs. 2000.0
Status: Within Budget

==================================== REST
Enter event budget: -5000
Enter venue expenses: 1000
Enter food expenses: 1000
Enter decoration expenses: 500
Invalid input
```